MIL-L-22189A(SHIPS)
18 July 1974
SUPERSEDING
MIL-L-22189(SHIPS)
5 April 1961
(See 6.4)

MILITARY SPECIFICATION

LIGHT, MARKER, AIRCRAFT, DITCHING

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a floating electric light complete with sea water activated batteries to provide emergency lighting in open sea for aircraft ditching at night and as an emergency seadrome lighting.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitaion for bids or request for proposal, forms a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-B-17482 - Batteries, Water Activated; And Batteries, Wet, Primary.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Ill. 60606).

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be examined and tested as specified in 4.2 (see 6.2).

3.2 Materials.

3.2.1 All materials used shall be of a first class quality and shall conform to the requirements of this specification (see figure 1).

3.2.2 All plastic materials shall be fully cured and essentially free from volatile plasticizers or fillers. Thermosetting plastics shall be used whenever practicable for parts under constant stress. Thermoplastic materials, whenever used, shall be of a formulation that has high-resistance to cold flow, shrinkage, and warpage or discoloration due to heat and sunlight. The use of any plastic that has a burning rate of more than 1.5 inches per minute shall not be employed.

3.2.3 All current-carrying parts shall be as specified and as shown on figure 1.

3.3 General requirements.

3.3.1 The light, when assembled with battery (see figure 1), shall be a self-contained unit for operation in the open sea, at water temperatures from 30° to 80° Fahrenheit (F).

FSC 6210

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3.3.2 Construction. The light shall be new and shall consist essentially of a lens, lamp housing, float structure, wire and battery holder. The light shall be watertight and vaporproof and shall comply with the requirements of this specification (see figure 1). The light complete with battery shall withstand the tests specified in 4.5.4.

3.3.3 Size. The light shall be cylindrical in shape. The dimensions of the light shall be as shown on figure 1.

3.3.4 Weight. The weight of the light and battery assembled shall not exceed 5-1/2 pounds.

3.4 Detail requirements.

3.4.1 Lamp housing. The lamp housing shall be either plastic or a light weight metal.

3.4.1.1 Plastic. If the lamp housing is of plastic material, it shall consist of either a tube of heavy paper base or other filler impregnated and bonded with a phenolic or a thermosetting resin or binder, known commercially as "XX" grade.

3.4.1.2 Metal. If the lamp housing is light weight metal, the wall thickness shall be not less than 0.025 inches.

3.4.2 Lens. The lens shall be made of a clear, colorless, transparent toughened glass or plastic material. It shall be reasonably free from bubbles, striae, checks, scratches and other optical or mechanical defects consistent with the candlepower requirements specified herein. The lens shall be fastened to the light by means of a bezel ring (see figure 1). The lens thickness shall be sufficient to withstand the drop test specified in 4.5.3. Lens shall have a smooth outer surface; the inner or outer surface shall be of a design which will provide for a ratio of peak beam candlepower of the light on the horizon to mean spherical candlepower of the lamp (bulb) of not less than five. The tests shall be conducted as specified in 4.5.5.

3.4.3 Battery. The light shall derive its source of electrical energy from one sea water activated battery in accordance with class N6C480, type I of MIL-B-17482. The battery shall not be furnished with the marker light.

3.4.4 Lamp. The lamp shall be new and of the bayonet candelabra type rated at 6-8 volts and 4.48 ampere (Industry No. 1133). The center of the lamp filament shall be at the focal point of the lens.

3.4.5 Socket. The socket shall be of corrosion-resistant metal, of durable construction and have adequate electrical clearances. Means shall be provided for minimizing the possibility of the lamp becoming loose due to vibration. The pressure exerted by the spring against the center contact of the lamp shall not be less than 30 nor more than 150 ounces.

3.4.6 Identification. The light shall be plainly and indelibly marked with the contract number, year of manufacture and the manufacturer's name or trademark.

3.4.7 Watertightness. The light shall be watertight and shall show no leakage of water into its interior when tested as specified in 4.5.3.

3.5 Workmanship.

3.5.1 Workmanship shall be first class in all respects. Lights shall be free from mechanical, electrical, or other imperfections or defects which materially affect appearances or which may affect serviceability.

3.5.2 All electrical connections, except those intended to be removable, shall be mechanically secured and soldered and shall not depend upon solder alone for mechanical strength. Where electrical connections are in close proximity, each shall be electrically insulated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5. Production shall not be started until the contractor has furnished evidence that the sample has passed the first article tests.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. For purposes of sampling, a lot shall consist of not more than 1,000 lights offered for delivery at one time.

4.3.2 Sampling for examination. A random sample of lights shall be selected from each lot in accordance with table I for the examination of 4.4.1.

Table I - Sampling for examination
AQL (approx.) = 1.5 percent defective.

Number of lights in lot	Number of lights in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 and over	75	3	4

4.3.3 Sampling for quality conformance test. A random sample of lights shall be selected in accordance with table II and shall be subjected to the test specified in 4.4.2.

Table II - Sampling for quality conformance test.

Number of lights in lot	Number of lights in sample
40 and under	3
41 to 110	5
111 to 300	7
301 to 500	10
501 to 800	15
801 and over	25

4.4 Quality conformance inspection.

4.4.1 Examination. Each of the sample lights selected in accordance with 4.3.2 shall be subjected to the examination specified in 4.5.1. Any lights in the sample containing one or more defects shall not be offered for delivery.

4.4.2 Quality conformance test. Each of the sample lights selected in accordance with 4.3.3 shall be subjected to the test specified in 4.5.2 to verify compliance with this specification. If any sample light fails to conform to this test the light and the lot represented by the light shall not be offered for delivery.

4.5 Inspection.

4.5.1 Examination. Lights shall be subjected to surface examination to ascertain that design, marking, material, weight, dimensions, shape, and workmanship are in conformance with the requirements of this specification. The fit of parts shall be observed with particular reference to the interchangeability of such parts as are likely to require replacement during the normal service life of the light.

4.5.2 General operation. Lights shall be fitted with battery and operated to determine compliance with the circuit requirements.

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4.5.3 Drop test. The light, complete with battery, shall be dropped twice from a height of 30 feet into sea water (or liquid of equal specific gravity). After each drop, the light shall be examined for water leakage, cracks or breakage. Failure of lamp filament shall constitute failure of the light.

4.5.4 Float test. The light, complete with battery shall be allowed to float for 1 hour in sea water (or liquid of equal specific gravity). At the beginning and end of this period the light center shall be not less than 10 inches above the surface of the water, and the light shall not have changed its waterline more than 1 inch.

4.5.5 Light output test. This test shall be made at a temperature of 70° to 80° F., after the battery has reached equilibrium, and within 30 days from the time of sampling, using the lamp supplied with the light. The ratio of candlepower of the light shall be measured for conformance to the requirements of 3.4.2.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, (see 6.3).

5.1 Packaging, packing and marking.

5.1.1 Packaging. Packaging shall be in accordance with the manufacturer's commercial practice.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, National stock number or manufacturer's part number, contract or order number, contractor's name and destination.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
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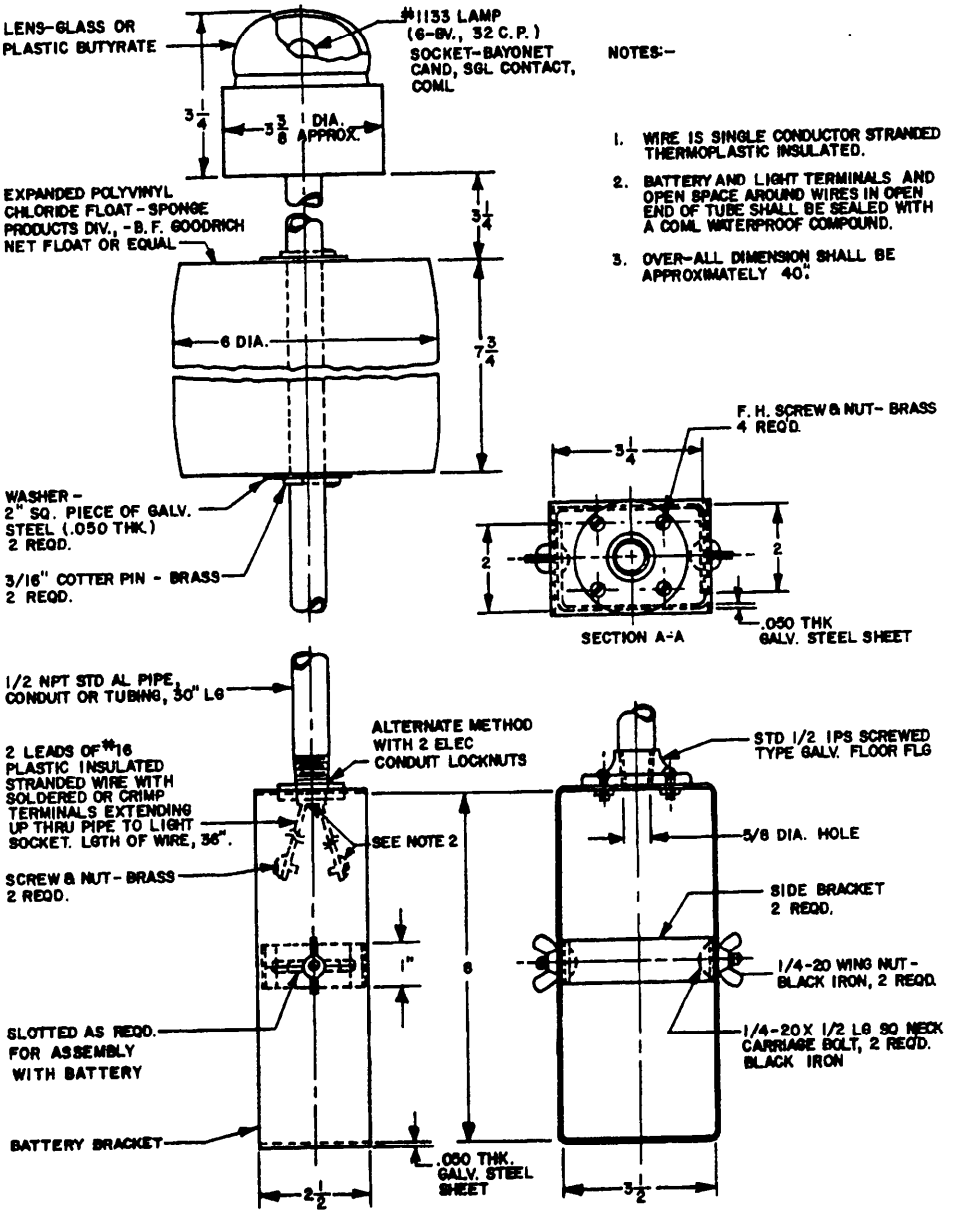


Figure 1 - Light, marker, aircraft ditching.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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